DATE:21/02/2026

EX-05

Stack based buffer overflow attacks

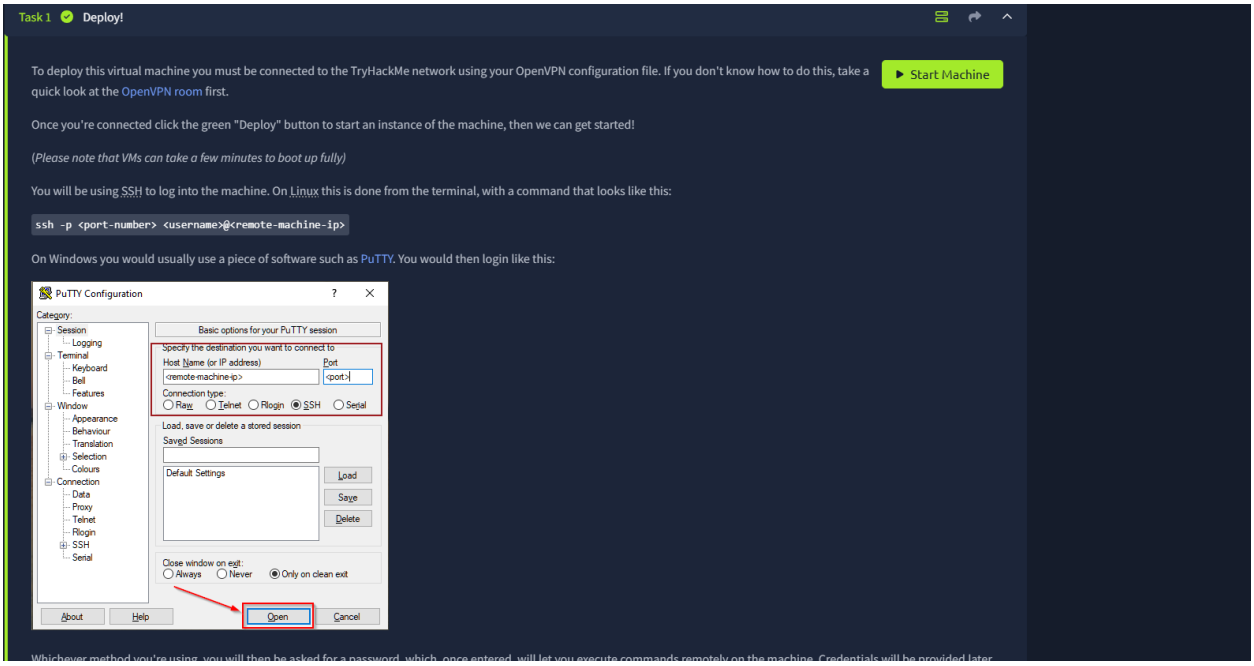
AIM:

To study the **Sudo buffer overflow vulnerability (CVE-2019-18634)** and understand how it can be exploited to gain **root privileges** on a Linux system.

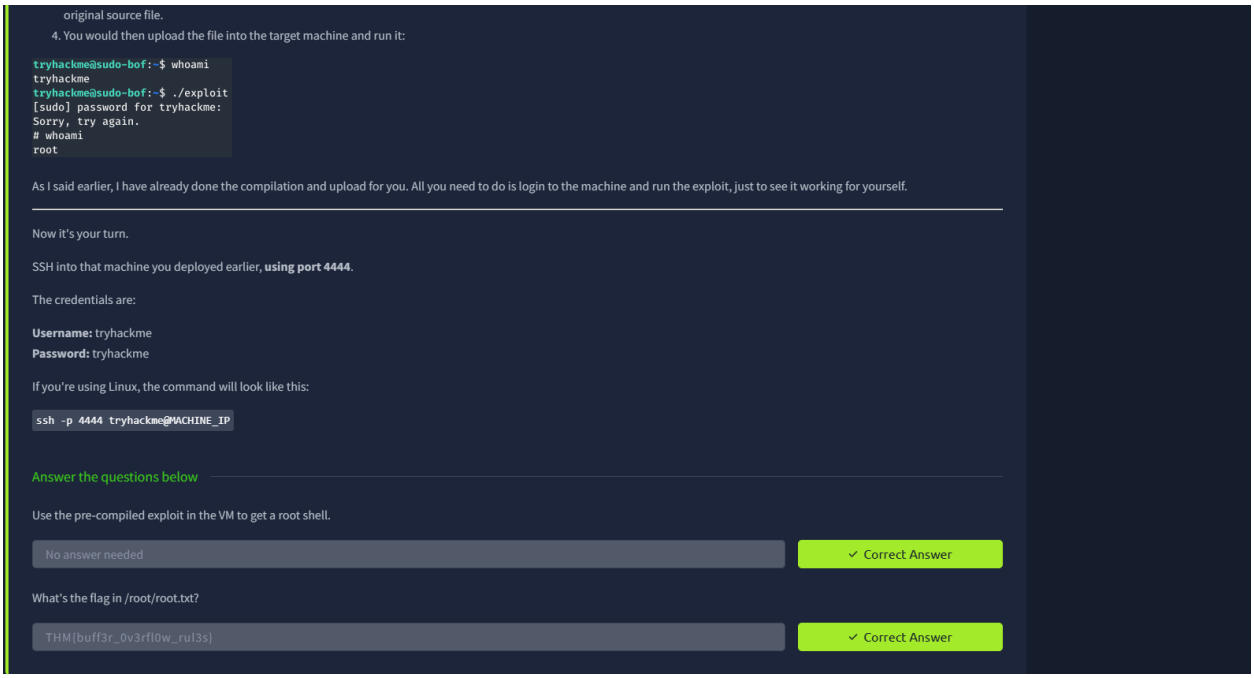
PROCEDURE:

- Start the TryHackMe machine and connect via SSH.
- Check the **sudo version** using `sudo -V`.
- Verify if **pwfeedback** is enabled in `/etc/sudoers`.
- Run the exploit to trigger the **buffer overflow**.
- Gain root access and capture the flag.

TASK-1:



TASK-2:



original source file.

4. You would then upload the file into the target machine and run it:

```
tryhackme@sudo-bof:~$ whoami
tryhackme
tryhackme@sudo-bof:~$ ./exploit
[sudo] password for tryhackme:
Sorry, try again.
# whoami
root
```

As I said earlier, I have already done the compilation and upload for you. All you need to do is login to the machine and run the exploit, just to see it working for yourself.

Now it's your turn.

SSH into that machine you deployed earlier, **using port 4444**.

The credentials are:

Username: tryhackme
Password: tryhackme

If you're using Linux, the command will look like this:

```
ssh -p 4444 tryhackme@MACHINE_IP
```

Answer the questions below

Use the pre-compiled exploit in the VM to get a root shell.

No answer needed — ✓ Correct Answer

What's the flag in /root/root.txt?

THM{buff3r_0v3rfl0w_rul3s} — ✓ Correct Answer

RESULT:

The vulnerability was successfully exploited, demonstrating that the **Sudo pwfeedback feature can cause a stack-based buffer overflow**, allowing a user to escalate privileges to **root**.